

fire prevention and control

fire prevention and control, the prevention, detection, and extinguishment of fires, including such secondary activities as research into the causes of fire, education of the public about fire hazards, and the maintenance and improvement of fire-fighting equipment.

Until after World War I little official attention was given to fire prevention, because most fire departments were concerned only with extinguishing fires. Since then most urban areas have established some form of a fire-prevention unit, the staff of which concentrates on such measures as heightening public awareness; incorporating fire-prevention measures in building design and in the design of machinery and the execution of industrial activity; reducing the potential sources of fire; and outfitting structures with such equipment as extinguishers and sprinkler systems to minimize the effects of fire.

The importance of increasing public understanding of the causes of fire and of learning effective reactions in the event of fire is essential to a successful fire-prevention program. To reduce the impact and possibility of fire, the building codes of most cities include fire safety regulations. Buildings are designed to separate and enclose areas, so that a fire will not spread; to incorporate fire-prevention devices, alarms, and exit signs; to isolate equipment and materials that could cause a fire or explode if exposed to fire; and to install fire-extinguishing equipment at regular intervals throughout a structure. Fire-retardant building materials have also been developed, such as the paints and chemicals used to coat and impregnate combustible materials, such as wood and fabric.

In the United States a study conducted over a 10-year period found that the most frequent type of fire was electrical (23 percent of all fires); other causes of fire included tobacco smoking (18 percent), heat caused by friction in industrial machinery (10 percent), overheated materials (8 percent), hot surfaces in such devices as boilers, stoves, and furnaces (7 percent), burner flames (7 percent), and combustion sparks (5 percent).

To reduce the hazardous effects of fire the most basic mechanism is an alarm system, which warns people to leave a building at once, alerts the fire department, and identifies the location of a fire within a structure. Besides the fire alarms that are triggered by people, there are many automatic devices that can detect the presence of fire. These include heat-sensitive devices, which are activated if a specific temperature is reached; a rate-of-rise detector, which is triggered either by a quick or a gradual escalation of temperature; and smoke detectors, which sense changes caused by the presence of smoke, in the intensity of light, in the refraction of light, or in the ionization of air.

Many public buildings are equipped with automatic sprinkler systems, which release a spray of water on an affected area if a fire is detected. The effectiveness of these systems has been proved in data accumulated from throughout the world: in buildings protected by sprinkler systems that had fires, the system extinguished fires in 65 percent of the cases and contained fires until other fire-fighting measures could be taken in 32 percent of the

cases. A major problem with sprinkler systems is the potential for water damage, but it has been found that in most cases this threat is minimal compared with the damage that a fire could cause.

There exists a considerable variety of fire-fighting equipment, ranging in sophistication from buckets and extinguishers to the elaborate yet portable apparatus used by fire departments. The most common of these is the fire engine, equipped with hoses, ladders, water tanks, and tools. Ladder and rescue trucks work in conjunction with trucks equipped with platforms that can be elevated by hydraulic lifts to carry out rescue efforts. Fireboats are employed in combating fires on ships and on waterfront property.

Fire-extinguishing agents other than water are used to fight various types of fire. Foaming agents are employed to handle oil fires. "Wet" water, formed by the addition of a chemical that reduces surface tension, can be used in a clinging foam to protect the exterior of a structure near the source of a fire. Ablative water, made by mixing water with additives, forms a dense, heat-absorbing blanket. Carbon dioxide is employed when water cannot be used and a fire must be fought by suffocation. Dry chemicals are used to extinguish electrical fires or burning liquids, while dry powder is utilized to put out such burning metals as magnesium and phosphorus. Halogenated hydrocarbons, commonly called halons, take the form of liquefied gas or vaporizing liquids at room temperature; they inhibit the flame chain reaction. Steam is used to control fire in confined areas, while inert gas is employed to extinguish gas, dust, and vapour fires.

Fire fighting is a battle against time. The initial priority is rescuing any occupants that may be in a burning building. Precedence is then given to any location from which the fire may spread to a neighbouring structure. A typical method of fire fighting is the over-and-under system. Working from inside the building, if possible, the bulk of the fire fighting takes place from below, while further attack is carried out from above in an effort to prevent the fire from spreading upward.

In rural areas water-tank trucks are usually needed, thus the time factor becomes even more critical. Bush, grass, and forest fires are frequently fought using the same equipment that is used on structural fires. Aircraft are sometimes employed to dump fire-retardant slurries or water mixtures on these blazes.

It has also become necessary to combat fires in pressurized chambers, including spacecraft. The combustion rate in these environments is much higher than it is under normal atmospheric pressure. Strict construction guidelines are followed to keep fire hazards to a minimum, and highly pressurized sprinklers are installed that act immediately upon any combustion.

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